# Mortadha Ghnimi

*AI/ML Engineer & Graduate Researcher | Computer Vision, RAG, ML Systems*

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## EDUCATION

**Grand Valley State University (GVSU) — M.S. in Artificial Intelligence (Expected Fall 2027)** 01/2026 – Present
Grand Rapids, MI

- Awarded a **merit-based Graduate Assistantship** supporting M.S. study and AI research at GVSU.
- **Relevant Coursework:** Machine Learning, Data Mining, Computer Vision, Generative AI, MLOps, Cloud Application Development, High-Performance Computing.

**University of Carthage, Faculty of Sciences of Bizerte — B.S. in Computer Science** 09/2022 – 06/2025
Bizerte, Tunisia

- **Ranked first across all undergraduate years** of the Computer Science bachelor's program.

## EXPERIENCE

**Graduate Research Assistant — Remote Sensing & Vegetation Monitoring** 01/2026 – Present
*Grand Valley State University* MI, USA

- Researching **vegetation-stress detection across 12,000 pixels in 4 parks**, using multispectral satellite time series to investigate scalable monitoring alongside drone observations.
- Built **Random Forest baselines and Presto-augmented models** to estimate expected NDVI and flag anomalies.
- Engineered **spatial and time-series data pipelines**, using 2022–2025 observations for training and validation and reserving a fully held-out 2026 test year for temporal evaluation.
- Developed **Google Earth Engine** scripts to export satellite observations and map pixel-level vegetation anomalies for investigation.

**AI & Automation Intern** 05/2026 – 08/2026
*Asjai Lou Casting* Remote / New York, NY

- Turned a static casting-industry book into a private digital platform with an authenticated member app, admin tools, and 181 directory records protected by collection-specific access controls.
- Implemented secure account activation with single-use keys, email verification, and transactional account creation; verified duplicate-order handling and refund revocation using synthetic orders.
- Built a Groq-powered RAG assistant with authorized retrieval, validated citations, and fallback handling; passed 65 automated checks and three hosted generation smoke tests.

**AI & Mobile Developer** 01/2025 – 12/2025
*Smart It Partner* Tunis, Tunisia

- Developed the **GymBox React Native app and BoxAI FastAPI backend**, integrating FAISS retrieval, Gemma-powered Q&A, and profile-based workout and meal recommendations.
- Built a **PyTorch computer-vision pipeline** for wildlife monitoring, classifying **10 forest animal species** with custom CNN architectures and data augmentation.
- Implemented **member screens, account flows, and shopping features**, connecting the app to backend APIs.

## PROJECTS

**1st Place Winner – Multiclass Image Classification Competition (Kaggle)**
*Python, TensorFlow, Keras, NumPy, Pandas*

- Won 1st place in Kaggle's 10C-W26 challenge with a 3-CNN ensemble (WideResNet, ResNet-110) trained from scratch.
- Achieved 93.4% competition test accuracy on 16×16 RGB images; implemented CutMix, Mixup, label smoothing, and cosine learning-rate decay.
- Combined 2 pseudo-labeling rounds at 0.95/0.98 confidence thresholds, 15 TTA passes per model, and weighted ensembling.

**GeoAgent — Agentic AI Research & Data Analysis Platform**
*Python, LangGraph, MCP, FastAPI, PostgreSQL, Docker*

- Built a research workbench with LangGraph, MCP tools, retrieval, Python analytics, and SQL to analyze datasets and produce cited briefs.
- Designed **plan–review–execute–verify** workflows with persistent state, human approval, recorded tool outputs, and source-linked reporting.
- Built a **50-task evaluation harness** to assess task completion, tool selection, grounding, latency, and inference cost.

**High-Performance Sparse Matrix Multiplication**
*C++, OpenMP, SIMD*

- Reached **#1 on the HPC leaderboard**, recording a **68.78× single-machine benchmark speedup** across six matrix widths with **6/6 correctness**.
- Optimized sparse matrix multiplication on **296 MB of transcriptomics data** using packed CSR, OpenMP, SIMD, cache tiling, and prefetching.

**Cost-Sensitive Healthcare Provider Fraud Detection**
*Python, Scikit-learn, Pandas, NumPy*

- Built a fraud-detection pipeline for **5,410 providers** with **74 engineered features** from inpatient, outpatient, and beneficiary data.
- Evaluated LR, Random Forest, Gradient Boosting, and MLP across **200 model fits**, reporting a best mean **ROC-AUC of 0.946**.
- Implemented cost-sensitive investigation rules balancing reimbursement-weighted losses from missed fraud against investigation costs.

## SKILLS

- **Languages:** Python, C++, Java, JavaScript/TypeScript, SQL
- **AI & Machine Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face, Computer Vision, Recommendation Systems
- **GenAI & Agents:** LLMs, RAG, LangChain, LangGraph, MCP, Tool Calling
- **Data & Geospatial:** Pandas, NumPy, Google Earth Engine, PostgreSQL, pgvector
- **Backend & Systems:** FastAPI, REST APIs, Docker, Git, OpenMP, SIMD